

Akshat Arya

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Professional Summary

Third-year CS student with research experience at Samsung on deepfake audio detection and a track record of shipping full-stack AI applications, from fine-tuned speech models to agentic workflow systems. Won the Notion AI Challenge 2026 (MLH) and reached the finals of the Meta PyTorch Hackathon out of 31,000+ teams. Seeking roles in ML engineering or AI systems development.

Experience

Samsung Research

May 2025 – January 2026

Deep Learning Research Intern (Part-time)

Remote, Bangalore

- Fine-tuned **RawNet2** and **LFCC-LCNN** models for **deepfake audio detection**, achieving **98.46% accuracy** and **0.00% EER** on 322 Kannada audio samples using **LFCC features**, **Sinc filterbanks**, and **Mel-scale initialization**.
- Compared **4 fine-tuning strategies** across both architectures, leveraging a **100K-parameter LFCC-LCNN (176x smaller** than RawNet2's 17.6M) with equivalent accuracy via **non-parametric LFCC extraction**, **MaxFeatureMap activation**, and **selective layer freezing**.

Bhatiyani Astute Intelligence

April 2025 – August 2025

IoT Research Intern (Part-time)

Remote, Bangalore

- Contributed to the development of an **offline IoT-based digital wallet** featuring **AES-256 encryption** and a **haptics-based interface** to improve accessibility for visually impaired users.
- Collaborated with the hardware team to implement **cryptographic modules** enabling **end-to-end encryption** for secure transactional data processing.

Projects

AdeptAI

May 2026 – June 2026

Production AI Assessment Platform (React, FastAPI, BullMQ)

Live | GitHub

- AI platform that transforms uploaded course material into rubric-aligned **assessments**, **question banks**, **presentation decks**, and **learning roadmaps** within multi-class collaborative workspaces.
- Async pipeline using **BullMQ** and **Redis** offloads long-running LLM workflows from the request path, streaming real-time job status via **Socket.io** with document parsing isolated in a dedicated **FastAPI** microservice.

CloudDash Multi-Agent Support Engine

February 2026 – April 2026

Multi-Agent RAG System (LangGraph, ChromaDB, FastAPI)

Live | GitHub

- Designed a **LangGraph** multi-agent system where a Triage agent classifies intent and orchestrates specialist agents (Billing, Technical, Escalation) through **stateful workflow transitions**, preserving context across handoffs.
- Hybrid retrieval combining **ChromaDB** vector search and **BM25** via **RRF**; layered guardrails for **PII masking**, prompt injection, and response validation with session-aware memory and pluggable **Redis**-backed persistence.

FileForge

October 2025 – December 2025

Cloud-Native File Processing Platform (FastAPI, Docker, RabbitMQ)

GitHub

- Event-driven platform where uploads trigger async workflows for transcoding, malware scanning, and **AES-256** encryption via **RabbitMQ** service orchestration, with **MinIO** for S3-compatible object storage.
- Independently deployable **FastAPI** microservices containerized with **Docker** and **CI/CD**-driven rolling deployments, enabling fault isolation and zero-downtime updates.

Education

R.V. College of Engineering

September 2023 – August 2027

Bachelor of Engineering in Computer Science | CGPA: 8.89/10.0

Bangalore, India

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, C, SQL, Bash
- **Web Development:** FastAPI, Node.js, Express.js, Flask, React.js, Tailwind CSS, Vite, Streamlit, REST APIs, JWT Authentication
- **AI & Machine Learning:** CNN, RNN, Transformers, LLMs, Gemini, LangChain, LangGraph, NLP, Computer Vision, TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, OpenCV
- **Databases & DevOps:** PostgreSQL, MongoDB, MySQL, Redis, RabbitMQ, MinIO, Docker Compose, NGINX, GitHub Actions, GCP, Git, Linux
- **Software Engineering:** OOP, Data Structures & Algorithms, Microservices Architecture, Asynchronous Processing, Agile Methodology

Achievements & Interests

- Won **Notion AI Challenge 2026 (MLH)** with **NotionOS**, an agentic workflow system integrating Notion with real-time task execution tools; awarded a \$500 cash prize and a one-to-one meeting with **Ivan Zhao**, CEO of Notion.
- Selected as a **finalist** for the **Meta PyTorch OpenEnv Hackathon x Scaler School of Technology**, ranking among the top **800 teams** out of **31,000+ registrations**.
- Selected for **Amazon ML Summer School 2026**, a highly selective program with approximately **3,000 participants** chosen from over **134,000 applicants** to learn core machine learning from Amazon scientists.